MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: TOPOGRAPHICAL SURVEY REPORT

Document Number: **MD-216-1000-EG-CV-TPD-1001**

Revision: D01

Revision Date: 10.10.2023

Status: AS BUILT

Outotec Document Number:	**900707**	**GAU01**	**ZZ01**	**CLA04**	**00004**	**OU100658894**	**3**
L&T Document Number:	**O19045**	**001**	**T**	**K**	**24757**	**001**	**3**

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	15.08.2019	FIRST ISSUE	A.M.F.C.E. / NZH	E. SIVA	PH	
A02	17.09.2019	COMMENTS INCORPORATED & ISSUED FOR INFORMATION	A.M.F.C.E. / NZH	E. SIVA	PH	
C01	06.02.2020	ISSUED FOR CONSTRUCTION	A.M.F.C.E. / NZH	E. SIVA	PH	
D01	10.10.2023	AS BUILT	A.M.F.C.E. / NZH	E. SIVA	PH	

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1. Introduction

The Topographical survey of the Mansourah-Massarah site located in an undeveloped desert area, in the central western region of Saudi Arabia, about 300 km northeast of Taif at the location of 22°42'21.54"N, 42°42'09.10"E covering an area of approximately 7.51 sq. kms. was performed by A.M Fazal Contracting Est. The fieldwork was carried out from July 14, 2019 to July 31, 2019. Processing of data & preparing drawings was carried out from August 1, 2019 to August 10, 2019 Topographical Survey and traverse was carried out as per scope of work using Trimble GPS instrument. Leveling of the new established benchmarks were done using Auto Level Leica (NA-2).

2. Survey Activities

Established Eleven (11) Temporary bench marks & Nineteen (19) Permanent bench marks. The activity started by establishing eleven (11) temporary bench marks named TP-1 to TP-11 along the proposed road as well as nineteen (19) permanent bench marks on locations given by the client. GPS Static observations were performed using the two (2) bench marks provided by the client, BM-14 & BM-15, as base reference points. Static data were processed with Trimble Software. Below are the lists of information used in processing the static data as well as the resulting coordinates of BM nos. MMGP-1 to MMGP-19 and TP-1 to TP-11.

Reference Permanent Benchmarks Table of Coordinates

BM: No	Easting	Northing	Elevation (in Meter)	Remarks
BM-14	262080.999	2525710.871	940.596	Rebar in concrete
BM-15	262647.441	2525197.156	938.043	Rebar in concrete

Permanent Bench Mark Established at MMGP Site

BM: No	Easting	Northing	Elevation (in Meter)	Remarks
MMGP-1	263814.120	2506370.437	915.444	Brass Plate in Concrete
MMGP-2	263777.124	2506328.659	915.310	Brass Plate in Concrete

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MMGP-3	263645.254	2506293.896	915.355	Brass Plate in Concrete
MMGP-4	263783.800	2506091.950	914.892	Brass Plate in Concrete
MMGP-5	263959.026	2505913.384	913.145	Brass Plate in Concrete
MMGP-6	263896.624	2506098.305	913.797	Brass Plate in Concrete
MMGP-7	264091.028	2506177.968	913.060	Brass Plate in Concrete
MMGP-8	264109.001	2505990.953	913.025	Brass Plate in Concrete
MMGP-9	264234.379	2505874.332	912.541	Brass Plate in Concrete
MMGP-10	264327.519	2506048.951	912.101	Brass Plate in Concrete
MMGP-11	264404.548	2506069.124	912.042	Brass Plate in Concrete
MMGP-12	264177.980	2506215.034	913.038	Brass Plate in Concrete
MMGP-13	264393.142	2506529.820	912.535	Brass Plate in Concrete
MMGP-14	264410.278	2506477.883	912.464	Brass Plate in Concrete
MMGP-15	263866.791	2506317.848	914.813	Brass Plate in Concrete
MMGP-16	264812.901	2505183.450	909.922	Brass Plate in Concrete
MMGP-17	264557.219	2504685.255	909.402	Brass Plate in Concrete
MMGP-18	263258.782	2508256.823	921.037	Brass Plate in Concrete
MMGP-19	263160.895	2508193.990	922.493	Brass Plate in Concrete

Temporary Benchmarks Table of Coordinates

BM: No	Easting	Northing	Elevation (in Meter)	Remarks
TP-1	261689.626	2523500.379	941.801	Rebar in concrete
TP-2	261092.953	2522117.675	947.704	Rebar in concrete
TP-3	260812.591	2520612.035	948.588	Rebar in concrete
TP-4	260658.747	2518078.879	942.268	Rebar in concrete
TP-5	260622.971	2516697.407	940.358	Rebar in concrete
TP-6	260587.696	2515665.713	941.331	Rebar in concrete
TP-7	260736.861	2514233.212	934.683	Rebar in concrete

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TP-8	260930.864	2512001.365	927.565	Rebar in concrete
TP-9	261343.202	2510258.456	926.934	Rebar in concrete
TP-10	262307.770	2508384.106	923.523	Rebar in concrete
TP-11	262837.767	2507514.021	919.070	Rebar in concrete

3. Coordinate System

Projection: UTM
Datum: WGS 1984
Zone: 38 North (45E)
Geoid: EGM08at1

4. GPS Topographical survey

The topographical survey was carried out using GPS instrument. Bench Mark number TP-11 was used as control point throughout the field work. The coordinates of the occupied stations were entered into the GPS instrument to initiate the survey work. Observations of spot heights and features both natural and man-made follow and were carried out as per scope of work.

5. Topo survey data processing and drawing preparations

The equipment has the capability to calculate and processed the observed data and export as an ASCII file or commonly called as text file (e.g. xyz, PENZD etc.)

The ASCII file(s) from the equipment were carefully verified and checked to make sure that the data are free from errors and then used in preparing the topographical drawing.

6. Software Used

1. Trimble Business Center (TBC)
2. SDR Mapping & Design
3. AutoCAD Civil 3D

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7. List of Attachments

1. Survey Record for Verification of Existing Bench Mark 14 & 15. (Attachment -1)
2. Topography Survey Report showing Contour. (Refer MD-216-1000-EG-CV-TPD-1002, A01).

8. Special Observation

The contour levels established in this topography survey report are approximately 5 meter above the levels indicated in BFS survey report (Ma'aden document no MD-216-0000-CP-CT-RFP-0111, Rev-C01) provided by Employer for survey area zones A,B,C,D,E,F,H & J. EPC confirms the present survey report as accurate based on repeat verification done at site. Accordingly this report will be considered as final topography survey report for this project and the BFS survey report document will be ignored.

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ATTACHMENT – 1
(Details of Bench Marks)

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VERIFICATION OF BELOW REFERENCE BENCH MARKS

Reference Document: **MD-212-0110-EG-CV-DEL-0001_D00**

Details of Reference Approved Bench Marks as mentioned in Reference Document -

Bench Mark Number	Northing	Easting	Elevation (m)
BM 14	2525710.871	262080.999	940.596
BM 15	2525197.156	262647.441	938.043

Actual Survey Readings at Site for the above Bench Marks:

Bench Mark Number	Northing	Easting	Elevation (m)
BM-14	2525710.866	262080.998	940.592
BM-15	2525197.156	262647.444	938.039

Project Specification for Topographical Survey : MD-101-SMEM-EG-CV-SPC-0008

Method Statement : MD-216-3000-EG-CV-GUI-0002

Instrument Details: -

- **Name** - TRIMBLE R4S

  NA-2

A. M. FAZAL KHAN CONST.	L&T	MA'ADEN
Name: MEHMOOD	Name: Jagan Suyambulingam	Name:
Signature:	Signature:	Signature:

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- **Make** – TRIMBLE

  LICA

- **Serial Number** – RS5EC98597

  459019

Calibration Details: - Refer Attachment I & II

Next Calibration Due: 03-Oct'19
12-Dec'19

Remarks, If any :

Photographs :

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Al-Jehat Company
Authorized Trimble Dealer

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JATCO
Trimble

شركة الجهات للتجارة والمقاولات
موزع معتمد ترمبل

Ref No: IAT-R19648
Aramco Vendor ID: 10013005

CALIBRATION CERTIFICATE

This is to certify that the corresponding unit,

Instrument: *GLOBAL POSITION SYSTEM*
Manufacturer: *TRIMBLE*
Model: *TRIMBLE R4s*
Serial No. *5847550174 5814550680*
Model: *TSC3, CONTROLLER w/TRIMBLE ACCESS*
Serial No. *RS5EC98597*

has been working propery to manufacturers specification and conform to it's standard and test were obtained using known base line and conventional methods to determine the error of such units. And results are within the tolerable limits.

OWNER: Ashraf Muhammad Fazal Khan Contractor Est.

Date: 4-Apr-19

Certified By:
AL JEHAT COMPANY (JATCO)

ص. ب ٢١٧٣ – الدمام ٣١٤٥١ – المملكة العربية السعودية – تلفون : ٨٣٣ ٦٦١٠ –١٣– ٩٦٦+ – فاكس : ٠٥٩٠ ٨٣٢ –١٣– ٩٦٦+
ص. ب ٥٣١٣٣ – الرياض ١١٥٨٣ – تلفون : ٤٤٥٤٩٧٧ –١١– ٩٦٦+ – فاكس : ٤٩٨٨ ٤٤٥ –١١– ٩٦٦+ – س. ت ٢٠٥٠٠٠٣٥٨٢
ص. ب ٨٠٦٢ – جدة ٢١٤٨٢ – تلفون : ٦٥٠ ١٦٤٠ –١٢– ٩٦٦+ – فاكس : ١٦٤٠ –٦٥٠ –١٢– ٩٦٦+
P. O. Box 2173 - Dammam 31451 - Saudi Arabia - Tel.: +966-13-833 6610 - Fax: +966-13-832 0590
P. O. Box 53133 - Riyadh 11583 - Tel.: +966-11-445 4977 - Fax: +966-11-445 4988 - C. R. 2050003582
P.O.Box 8062 - Jeddah 21482 - Tel.: +966-12-650 1640 - Fax: 650 1640
E-mail: trimb@jatco.com.sa : بريد الكتروني

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شركة الجهات للتجارة و المقاولات
موزع معتمد ترمبل

Ref No: [redacted]
Aramco Vendor ID: [redacted]

CALIBRATION CERTIFICATE

This is to certify that the corresponding unit,

Instrument:	GLOBAL POSITION SYSTEM
Manufacturer:	TRIMBLE
Model:	TRIMBLE R4s
Serial No.	5812550867
Model:	TSC3, CONTROLLER w/TRIMBLE ACCESS
Serial No.	RS5DC98024

has been working propery to manufacturers specification and conform to it's standard and test were obtained using known base line and conventional methods to determine the error of such units. And results are within the tolerable limits.

OWNER: Ashraf Muhammad Fazal Khan Contractor Est.

Date: 4-Apr-19

Certified By:
AL JEHAT COMPANY (JATCO)

P. O. Box 2173 - Dammam 31451 - Saudi Arabia - Tel.: +966-13-833 6610 - Fax: +966-13-832 0590
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ZAN MODERN CO. LTD.
شركة زان الحديثة المحدودة

VAT No. 300127295900003 الرقم الضريبي ٣٠٠١٢٧٢٩٥٩٠٠٠٣ C.R.: 4030192824 س.ت: ٤٠٣٠١٩٢٨٢٤

Calibration Certificate

Customer : محمود يول حق هاشم

Instrument type : *Automatic Level*
Model : *NA-2*
Serial Number : 459019

We hereby certify that the following survey instrument has been calibrated to Topcon Japan standards conditions in Authorized Topcon service center.

The instrument is calibrated within the tolerance.
The calibration was performed under conditions stated.

Temperature : 24 ° C
Relative Humidity : 40 %

Date : 12–06–2019
Due Date : 12–12–2019

Service Engineer

Eng . Gamal Hussein

Head Office: Jeddah - Madeena Cross Road with king abdullah - Tel.: 012 6576883/4 Fax: 012 6576881 المركز الرئيسي: جدة – تقاطع طريق المدينة مع الملك عبدالله – ت: ٠١٢ ٦٥٧٦٨٨٣/٤ فاكس: ٠١٢ ٦٥٧٦٨٨١

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